

Our objective is to figure out a way to write $f(x)$ as a Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx.$$

First, we should figure out how to find a_k, b_k .
Many of you have learned dumb formulas for this that don't generalize.

Our goal is to make this a linear algebra problem. In finite dimensions, given an n dimensional vector space V with an inner product $\vec{v} \cdot \vec{w}$, we say $\vec{v} \perp \vec{w}$ if $\vec{v} \cdot \vec{w} = 0$. An orthonormal basis for V is a collection of vectors $\vec{u}_1, \dots, \vec{u}_n$ so that

$$\begin{aligned} \vec{u}_i \cdot \vec{u}_j &= 0 & i \neq j \\ &= 1 & i = j \end{aligned}$$

and each $v \in V$ can be written

$$\vec{v} = c_1 \vec{u}_1 + \dots + c_n \vec{u}_n.$$

Importantly, the coefficients c_i are extremely easy to compute:

$$c_i = \vec{v} \cdot \vec{u}_i.$$

so given o.n. basis $\{\vec{u}_i\}$, every $\vec{v} \in V$ can be expressed

$$\vec{v} = \sum_{k=1}^n (\vec{v} \cdot \vec{u}_k) \vec{u}_k.$$

So we'd like to construct a vector space of functions with $\cos kx, \sin kx$ as an orthonormal basis.

Def: Given functions f, g on $[-\pi, \pi]$, the L^2 inner product $\langle f, g \rangle$ is given by

$$\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) g(x) dx.$$

the L^2 -norm of f is

$$\|f\| = \sqrt{\langle f, f \rangle}$$

$$\text{or } \|f\|^2 = \langle f, f \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)^2 dx.$$

You might show that this is indeed an inner product.

Lemma:

$$\begin{aligned} \langle \cos kx, \cos lx \rangle &= 0 & k \neq l \\ \langle \sin kx, \sin lx \rangle &= 0 & k \neq l \\ \langle \cos kx, \sin lx \rangle &= 0 & \text{for all } k, l. \end{aligned}$$

$$\|1\| = \sqrt{2}$$

$$\|\cos kx\| = 1 \quad \text{for all } k$$

$$\|\sin kx\| = 1 \quad \text{for all } k.$$

That is, $1, \cos x, \sin x, \cos 2x, \sin 2x, \dots$

is an orthogonal system of functions.

How can we use this to compute coefficients?

Suppose $f(x) = \frac{a_0}{2} + \sum_k a_k \cos kx + b_k \sin kx.$

then $\langle f, \cos lx \rangle$

$$= \left\langle \frac{a_0}{2} + \sum_k a_k \cos kx + b_k \sin kx, \cos lx \right\rangle$$

$$= \cancel{\left\langle \frac{a_0}{2}, \cos lx \right\rangle} + \sum_k a_k \underbrace{\langle \cos kx, \cos lx \rangle}_{\substack{0 \\ \text{unless } k=l}} + b_k \cancel{\langle \sin kx, \cos lx \rangle}_0$$

$$= a_l \langle \cos lx, \cos lx \rangle$$

$$= \underline{a_l}.$$

$$\text{so } a_k = \langle f, \cos kx \rangle.$$

$$\text{likewise, } b_k = \langle f, \sin kx \rangle$$

$$a_0 = \langle f, 1 \rangle \quad (\text{lemma 2}).$$

$$\text{so } \boxed{f(x)} = \frac{a_0}{2} + \sum_k a_k \cos kx + b_k \sin kx$$

$$\text{then } f(x) = \frac{1}{2} \langle f, 1 \rangle + \sum_k \langle f, \cos kx \rangle \cos kx + \langle f, \sin kx \rangle \sin kx.$$

lets compute one, leaving aside questions of convergence for now.

Ex: let $f(x) = x$ on $[-\pi, \pi]$.

$$\langle x, 1 \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} x dx = \frac{1}{\pi} x^2 \Big|_{-\pi}^{\pi} = \frac{1}{\pi} (\pi^2 - (-\pi^2)) = 0.$$

$$\langle x, \cos kx \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos kx dx$$

$$\begin{array}{l} x \cos kx \\ 1 \quad \frac{1}{k} \sin kx \\ 0 \quad -\frac{1}{k^2} \cos kx \end{array}$$

$$\begin{aligned} & \left(\frac{1}{k} x \sin kx + \frac{1}{k^2} \cos kx \right) \Big|_{-\pi}^{\pi} \\ &= \frac{1}{\pi} \left((0 + (-1)^k) - (0 + (-1)^k) \right) = 0. \end{aligned}$$

$$\langle x, \sin kx \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin kx dx$$

$$= \frac{1}{\pi} \left(-\frac{x}{k} \cos kx + \frac{1}{k^2} \sin kx \right) \Big|_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left(\left(-\frac{\pi}{k} (-1)^k + 0 \right) - \left(\frac{\pi}{k} (-1)^k \right) \right)$$

$$= -\frac{2}{k} (-1)^k$$

$$= \frac{2}{k} (-1)^{k+1}$$

So $x \sim \sum_{k=1}^{\infty} (-1)^{k+1} \frac{2}{k} \sin kx.$

$$= 2 \sin x - \frac{2 \sin 2x}{2} + \frac{2 \sin 3x}{3} - \frac{2 \sin 4x}{4} + \dots$$

convergence is tricky. $\sim$ means "has the Fourier representation"

So what are we doing here? basically projecting f onto the span of $1, \cos x, \sin x, \cos 2x, \sin 2x, \dots$

A Fourier series is not a new object like a Taylor series.

If $f(x) = \sum_{n=0}^{\infty} a_n x^n$, and it converges, automatically

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \text{ and so on.}$$

what about our example?

$$x \sim \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} 2 \sin(nx)$$

$\frac{d}{dx}$

1 ? $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \cdot 2 \cos(nx)$

nup

$= \sum_{n=1}^{\infty} (-1)^{n+1} 2 \cos(nx)$. uh oh...

Another example of the flexibility of Fourier series:

Consider $\delta(x)$, the Dirac delta "function".

$$\delta(x) = 0 \quad x \neq 0$$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1.$$

Evaluation Property:

$$\int_{-\infty}^{\infty} f(x) \delta(x) dx = f(0).$$

Fourier Series

$$\langle \delta, \cos kx \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos kx \delta(x) dx = \frac{1}{\pi} (1)$$

$$\langle \delta, \sin kx \rangle = 0$$

$$\langle \delta, 1 \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} \delta(x) dx = \frac{1}{\pi}.$$

$$\text{so } \delta(x) \sim \frac{1}{2\pi} + \sum_{k=1}^{\infty} \frac{1}{\pi} \cos(kx)$$

So we have coefficients, the function obviously doesn't seem to converge pointwise, and δ isn't even a function.

we need to start developing a precise description of convergence.

periodic extension:

Since $\sin(kx)$ and $\cos(kx)$ are all periodic of period 2π , the limiting function, if it exists, must also be period 2π .

$$\tilde{f}(x+2\pi) = \tilde{f}(x) \quad \text{for all } x \in \mathbb{R}.$$

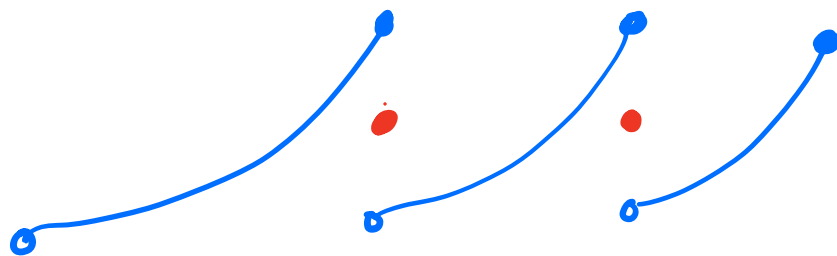
Lemma:

Periodic Extension:

Suppose f is defined on $(-\pi, \pi]$.

then there is a unique 2π -periodic function $\tilde{f}$

so that $\tilde{f}(x) = f(x)$ on $(-\pi, \pi]$.



we've already seen that

$$x \sim 2 \underbrace{\sum_{k=1}^{\infty} \frac{(-1)^{k+1} \sin kx}{k}}_{f(x)}.$$

in fact, $f(x)$ can be shown to converge to the Fourier periodic extension of x .

$$\text{that is } 2 \sum_{k=1}^{\infty} (-1)^{k+1} \frac{\sin kx}{k} = \begin{cases} x & x \in (-\pi, \pi) \\ 0 & x = -\pi, \pi. \end{cases}$$

lots of these series gave number theoretic insights.

$$\underline{\text{ex}} \quad 2 \sum_{k=1}^{\infty} (-1)^{k+1} \frac{\sin(k \frac{\pi}{2})}{k} = \frac{\pi}{2}$$

$$= 2 \left(1 - 0 - \frac{1}{3} + 0 + \frac{1}{5} - 0 - \frac{1}{7} + \dots \right) = \frac{\pi}{2}$$

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$$

piecewise continuous function.

Def: A function is piecewise cts on $[a, b]$ if it is continuous except at a finite number of points $a \leq x_1 \leq x_2 \leq \dots \leq x_n \leq b$. AND the one sided limits

$$f(x_k^-) = \lim_{x \rightarrow x_k^-} f(x) \quad f(x_k^+) = \lim_{x \rightarrow x_k^+} f(x) \text{ exist.}$$

the points $x_1, \dots, x_k$ are called jump discontinuities.

Define $\beta_k = f(x_k^+) - f(x_k^-)$ which is the magnitude of the jump.

If $\beta_k = 0$, the discontinuity is removable.

The unit step function

$$\sigma(x) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases}$$

is piecewise cts. $\beta_0 = 1$. $\infty \sigma(0^+) - \sigma(0^-)$
 $= 1 - 0 = 1.$

More generally $h(x) = \beta \sigma(x - \xi)$ is a step function w jump magnitude β at $x = \xi$.

Any piecewise continuous function has well-defined FOURIER COEFFS, BUT may not converge.

Def: a function is piecewise C^1 on $[a, b]$ if it is defined, etc, and continuously differentiable except at $x_1, \dots, x_n \in [a, b]$ where $f(x_k^+)$, $f(x_k^-)$, $f'(x_k^+)$, $f'(x_k^-)$ exist.

for piecewise C^1 functions, there are two classes of exceptional points.

jump discontinuities ($f(x_k^+) \neq f(x_k^-)$)

corners ($f(x_k^+) = f(x_k^-)$ but $f'(x_k^+) \neq f'(x_k^-)$)

Theorem: If $\tilde{f}(x)$ is a 2π -periodic, piecewise C^1 function, then for any $x \in \mathbb{R}$, its Fourier series converges to

$$\begin{aligned} &\tilde{f}(x) \quad \text{if } \tilde{f} \text{ is continuous at } x \\ &\frac{1}{2} (\tilde{f}(x^+) + \tilde{f}(x^-)) \quad \text{if } x \text{ is a jump discontinuity.} \end{aligned}$$

Example: $\sigma(x)$.

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} \sigma(x) dx = \frac{1}{\pi} \int_0^{\pi} dx = 1$$

$$a_k = \langle \sigma(x), \cos(kx) \rangle = \frac{1}{\pi} \int_0^{\pi} \cos(kx) dx = 0$$

$$b_k = \langle \sigma(x), \sin(kx) \rangle = \frac{1}{\pi} \int_0^{\pi} \sin(kx) dx = \begin{cases} \frac{2}{k\pi} & k=2l+1 \\ 0 & k=2l \end{cases}$$

$$\text{so } \sigma(x) \sim \frac{1}{2} + \frac{2}{\pi} \sum_{k \text{ odd}} \frac{\sin(kx)}{k}$$

$$\begin{aligned} \tilde{f}(x) &= 0 & (2m-1)\pi < x < 2m\pi \\ &1 & (2m)\pi \leq x < (2m+1)\pi \\ &\frac{1}{2} & x = m\pi. \end{aligned}$$

Read Section on even and odd functions.